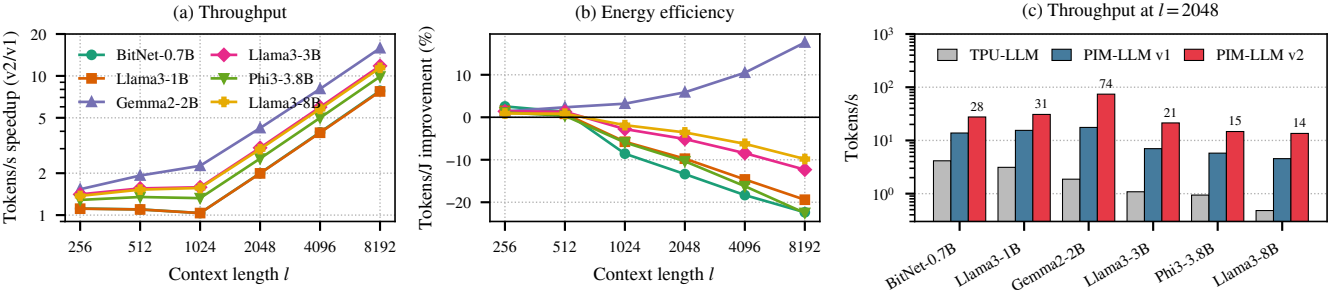
(a) Throughput
Tokens/s speedup (v2/v1)
20
10
5
2
1
BitNet-0.7B
Llama3-1B
Gemma2-2B
Llama3-3B
Phi3-3.8B
Llama3-8B
256
512
1024
2048
4096
8192
Context length $l$
(b) Energy efficiency
Tokens/J improvement (%)
10
0
−10
−20
256
512
1024
2048
4096
8192
Context length $l$
(c) Throughput at $l = 2048$
Tokens/s
$10^3$
$10^2$
$10^1$
$10^0$
TPU-LLM
PIM-LLM v1
PIM-LLM v2
28
31
74
21
15
14
BitNet-0.7B
Llama3-1B
Gemma2-2B
Llama3-3B
Phi3-3.8B
Llama3-8B